Minal Arunashalam

U.S. Citizen | Aldie, VA | 703-667-0316 | minal-dev.vercel.app | minal.shalam@gmail.com | Linkedin | Github

Technical Skills

Languages: Python, Java, C, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks: PyTorch, TensorFlow, scikit-learn, NumPy, OpenCV, Flask, React, FastAPI, LangChain, LangGraph
Tools: Git, Docker, Kubernetes, Terraform, Spark, MLFlow, MongoDB, PostgreSQL, AWS, GCP, Databricks, Claude Code, Cursor

Experience

AI/ML Engineer, StrategyCorps — **February 2026 – Present**

- Developing bank-specific LightGBM propensity models with Platt scaling on customer behavior and transaction data, enabling nearly 400 banks and credit unions to target customers for product-adoption campaigns across millions of daily records.
- Built transaction tagging NLP models with full MLOps setup in Databricks (MLFlow, drift detection and retraining using LLM pseudo-labeling, human-in-the-loop step, and automatic champion-challenger model deployment), achieving 99% classification accuracy on 1k+ merchants.
- Designed and built an LLM pipeline that classifies uncategorized product and transaction codes from core banking data, speeding up FI onboarding and reducing manual workload by 90%.
- Designed a conversational analytics agent using LangGraph, Databricks MCP, and Qdrant/Azure for RAG, deployed via MLFlow with a Vercel AI SDK frontend.

ML Engineer Intern, StrategyCorps — **September 2025 – Dec 2025**

- Designed and built a production MLOps pipeline with Databricks and MLFlow that serves 300+ banks, with automated training, deployment, and monitoring of ML models with 99% uptime and scalable support for millions of records daily.
- Trained and evaluated an XGBoost model for product adoption with class-imbalance handling and SHAP explainability.

AI Engineer Intern, Connected Things — **June 2025 – Aug 2025**

- Implemented graph neural networks (GraphSAGE, R-GCN), Autoencoder, XGBoost, and TransE models for link prediction on Wikidata, evaluated using MRR and Hits@10.
- Automated API documentation with LLM system (LangGraph, Gemini, FastAPI, Celery, and Redis), reducing manual documentation effort by 95% for developers.
- Deployed system on AWS with SQS + Lambda, ensuring 99.9% uptime and securing EC2/Redis in private VPC.

Teaching Assistant, GMU - College of Engineering — **Sept 2023 – May 2025**

- Instructed 2,000+ students across 4 semesters in Python, Java, and C, emphasizing OOP and test-driven development.

Lead Software Engineer, US Naval Research Lab (GMU Partnership) — **Sept 2024 – May 2025**

- Engineered 3D drone & UUV dynamics in Pyquaticus, enabling multi-axis DRL agent training for naval simulation research.
- Created 200+ unit/integration tests, achieving full coverage of drone/UUV dynamics and ensuring reliability.

Machine Learning Research Assistant, Human-Centric Design Lab – GMU — **Jan 2024 – Nov 2024**

- Quantized CNN model to TensorFlow Lite, achieving ~95% accuracy with <50 ms inference latency.
- Automated deployment of FastAPI + Docker model on AWS EC2 with Prometheus/Grafana monitoring and GitHub Actions updates.

Education

George Mason University — **May 2026**
Master of Science, Computer Science - Machine Learning Concentration — GPA: 3.67

- **Relevant Coursework:** Advanced NLP, Advanced ML, Deep Learning, Component-based Software Development (DevOps, Kubernetes, Terraform, Ansible, Spring Boot, AWS, etc.)

George Mason University - Honors College — **May 2025**
Bachelor of Science, Computer Science — GPA: Cum Laude

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Databases, Low-Level Programming, Networking, Object-Oriented Programming

The University of Texas at Austin — **Feb 2025**
Certificate in Artificial Intelligence and Machine Learning — GPA: 3.89/4.0

- **Relevant Coursework:** Generative AI, Computer Vision, Model Deployment, Recommendation Systems, Big Data Analytics (Apache Spark, Kafka), AI on Cloud (AWS Sagemaker)